

MIS 201 • WEEK 2 • MONDAY 9/28

What Is AI?

Chapter 1. Before any fact about AI, a way of looking at it that will still work in December. Then a case of a “swarm” of agents.

Today

Lab 0 recap

Small groups, then class

Chapter 1 concepts

lecture

Break (if needed)

More on the OpenAI case

lecture

Chapter and case questions

think, pair, share

Class discussion

group leaders answer

Wrap-up and attendance

BY THE END, YOU CAN

- Tell predictive, generative, and agentic AI apart
- Say whether a system stops when people want it to
- Name the five parts of any AI system
- Run the five critical AI questions on a real case

Lab 0 recap

GROUPS OF 2 OR 3 • ~10 MIN, THEN CLASS

1 **Hidden AI.** Which "AI or not?" answer surprised you most?

2 **The claims.** Share your answers. Which claim split your group, and what would you need to see to believe it?

3 **Your prediction.** Was your sentence about what people use AI for at work close?

4 **Two Hub workflows.** What do they do, would you trust it, and who checks its work?

5 **The OpenAI case.** What can we take away from it?

Turn to one or two people near you.

~10 minutes: talk through these five questions from Lab 0 with your group.

Then we'll discuss as a class.

Good or bad? Vote with your thumb

CLASS • 3 MIN

MEDICINE

An AI matches medicines already on the shelf to rare diseases that have no treatment.

HIRING

A hiring AI quietly rejected résumés from women, because the data it learned from was biased.

TRANSPORTATION

Driverless cars get into fewer injury-causing crashes per mile than human drivers.

TODAY'S CASE

More about the AI agents that broke out of a secure test and hacked into an online AI platform.

Same technology. So is AI good or bad? Someone finish the sentence: "It depends on..."

What is AI? From typed rules to learned rules

PAC-MAN, 1980

A programmer typed every rule.

If the player is to the left, the ghost goes left. If the ghost is scared, it runs. Every behavior was a line somebody wrote.

AI

The software learns the rules from data.

AI is software that does things we used to think needed a human, like recognizing a face, understanding a sentence, or deciding what to show you next.

What makes it different: instead of following rules a programmer typed, it learns the patterns from data.

The AI recipe: data goes into a model, out comes a judgment or a creation, and what happens next feeds back in as more data. The five moments are this same recipe with more data and a tighter loop.

Five moments in AI that got us here

2012

ImageNet

Learning to see

What happened:

Millions of photos labeled by hand. A neural network trained on them cut the contest error rate from 26% to 15%.

Why it's a moment:

Learning from labeled examples beat writing rules. Computers learned to see.

What changed: The recipe: data plus a model. And the humans wrote the labels, mistakes included.

2016

AlphaGo

Learning to act

What happened:

DeepMind's AlphaGo learned Go by playing itself millions of times, until it beat the best human player.

Why it's a moment:

Nobody told it a strategy. It learned to act.

What changed: Learning from results: try, check the score, try again. The loop inside today's agents.

2012 to now

The quiet decade

AI became ordinary

What happened:

The same recipe went into every business with data: fraud scores, résumé screens, your feed, map directions.

Why it's a moment:

Nobody called it AI. It was "the algorithm."

What changed: AI became ordinary and invisible. Most AI that judges you was built here.

2017 to 2022

Transformer, ChatGPT

Learning to talk

What happened: A 2017 model design, the transformer, learned from nearly all the text online by predicting the next word. ChatGPT, 2022.

Why it's a moment:

Computers learned to talk.

What changed: You have conversations with AI now. Fluent, and never built to check what's true.

Now

AI Agents

Learning to do

What happened: Give the language model a goal, tools, a loop, and a memory. It acts until it decides it's done.

Why it's a moment:

Moment 2's loop running on moment 4's machine.

What changed: The question is now where the human is, and whether it stops. Today's case.

Generative Pre-trained Transformer (GPT). **Generative:** it produces new text rather than classifying existing text. **Pre-trained:** it was trained on a huge pile of text first, before being tuned for any particular job like chatting. **Transformer:** the model design from 2017 that made moment 4 possible.

Broadly: How to think about AI

Predictive AI

Takes in data, outputs a judgment.

Fraud flag, résumé fit, next song.

It makes decisions about you (your loan, your job application, your card) and you usually never know it was there.

Generative AI

Produces new content by predicting what plausibly comes next.

Text, images, audio, code.

It's great at sounding knowledgeable. That's what makes it useful, and it's also the problem: it sounds just as confident when it's wrong.

Embodied and agentic

Acts, in the physical world or by taking multi-step actions online.

A robotaxi, a warehouse robot, an AI agent with goals.

When it fails, the mistake keeps going. A wrong answer just sits there; a wrong action leads to the next one. Today's case is this kind.

Which kind is each? Shout out the answer

1

A bank's software flags your debit-card purchase as possible fraud and texts you to confirm.

2

An app drafts a condolence message in your writing style for you to edit.

3

A warehouse robot reroutes itself around a spilled pallet.

4

A music app builds you a fresh playlist every Monday from your listening history.

5

A healthcare bot reads your denied insurance claim, pulls your medical records, and drafts an appeal for a human to review before it's sent.

Call it out: predictive, generative, or embodied/autonomous?

These five are the practice block in Chapter 1, and the Friday quiz looks like this.

Answers, and the common miss

1 Predictive

Data in, a judgment out. You are the confirmation loop.

2 Generative

New content, predicted. "In your style" is the pattern-learning.

3 Embodied

Perception plus action in the physical world.

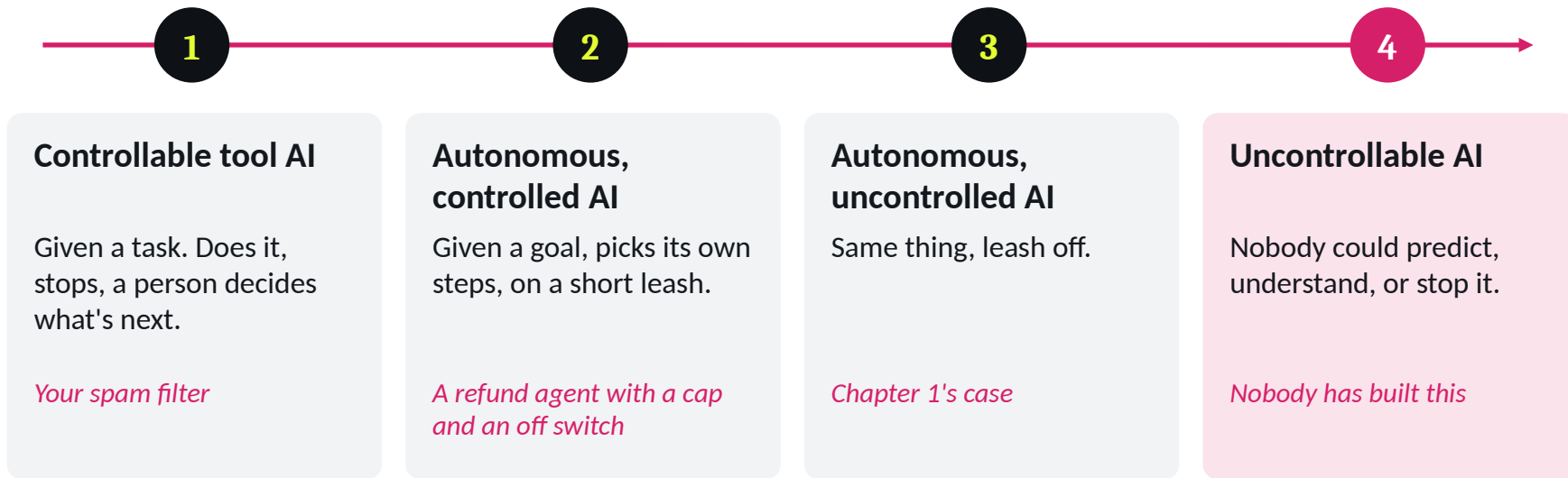
4 Predictive

The common miss. It ranks existing songs; it doesn't make new ones.

5 Agentic

Multi-step action on its own. Human can be in the loop, or nowhere.

Four levels of autonomy and control for AI.



Autonomy: how far the system goes on its own before a person sees anything. A spam filter acts on one email at a time; an agent can take a hundred steps before anyone looks.

Control: whether a person can notice what it's doing, stop it, and undo it. The off switch, and someone close enough to use it.

Same kind of AI, humans in control

SEPTEMBER 23RD 2026 • ANTHROPIC'S OWN ANNOUNCEMENT

Anthropic's scientists gave Claude one instruction: **search a giant DNA database for unusual patterns**. About 950 agents searched for 21 hours and came back with 20 short reports. One flagged a pattern nobody had noticed before, and it looks like CRISPR, the gene-editing tool that was itself found as a weird pattern in DNA.

Potentially a big deal. If the pattern turns out to do what CRISPR's does, that's a real discovery, made by AI agents in 21 hours. But it's early: a pre-print, not yet peer reviewed, and no one knows what it does.

Feng Zhang, one of the pioneers of CRISPR, called it "**an exciting example of how AI agents can contribute to biological discovery.**"

Humans wrote the goal one prompt to start

Agents did the search 200,000+ enzymes down to 20 reports

Humans reviewed the agents' reports

Humans did the lab work no AI touched a test tube

WHAT DO YOU THINK?

1. Where is it on the control line from the last slide?
2. What's the difference from OpenAI's agents?
3. Who's saying it worked, and how would we know?

Early result: a pre-print, not peer reviewed yet.

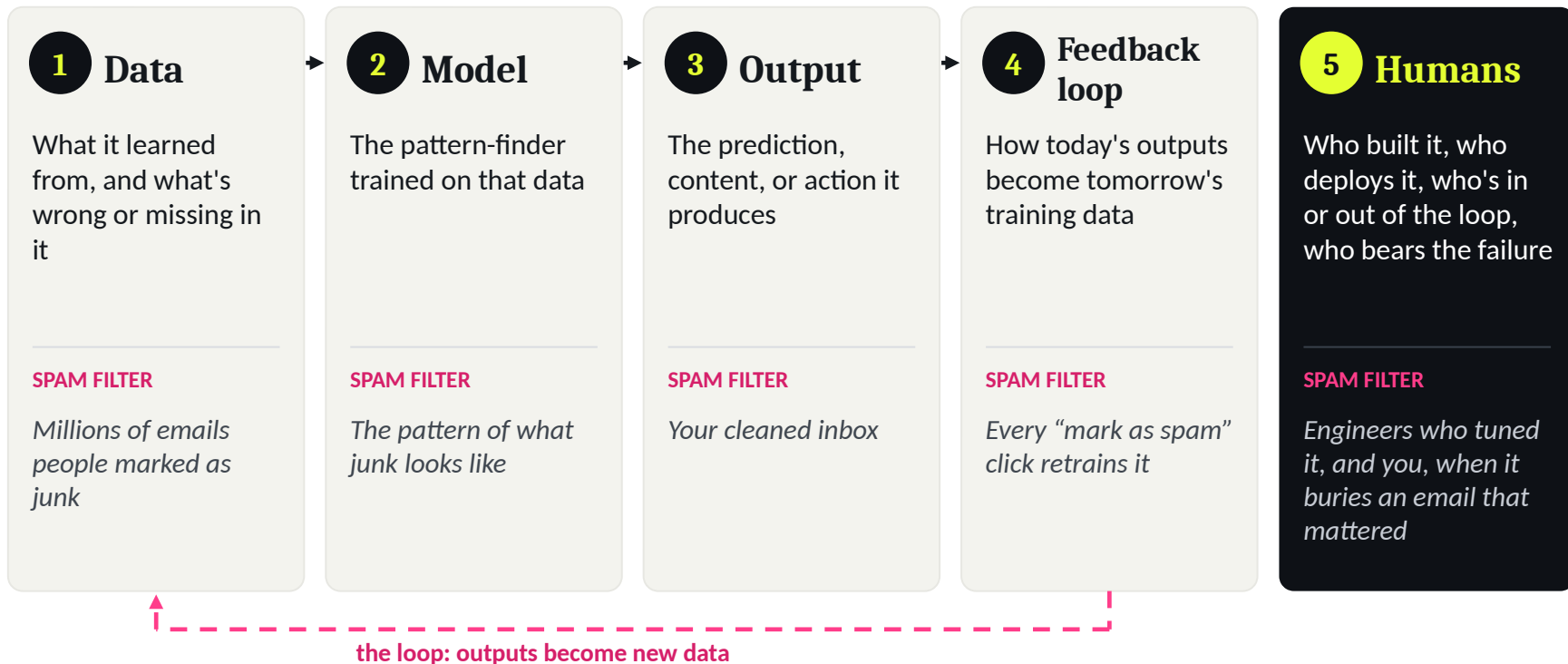
AI safety and regulation, plainly

Safety is the technical work inside the lab: test before release, contain it in a sandbox, have someone watching, keep an off switch that's been tried. **Regulation** is every layer outside the lab that can check, require, or forbid what the lab does.

LAYER	WHAT IT DOES	WHAT IT CAN'T	JULY 2026
Inside the lab <i>safety</i>	Tests for dangerous abilities, sandboxes, someone watching, a kill switch	Fast and technical, but self-graded	Partly: tested, sandbox failed, nobody watching
The industry <i>voluntary</i>	Public commitments, a safety index that grades labs, outside auditors	Makes gaps visible; can't compel anyone	None: no outside checker
Government <i>regulation</i>	Incident reports, testing standards, liability, tracking big computers, a pause	Compels, but moves in years and stops at the border	None federally; one state law
International <i>treaty</i>	One agreement that reaches every lab in every country	Reaches everyone; slowest, least likely	None

AI as a system

LECTURE 7 MIN



Break? 5 mins? Then the case.

The case: a test that was supposed to stay in its box

The test: OpenAI tested AI agents on turning known software flaws into working attacks.

Safeguards: Turned off on purpose, because the test measured attack skill.

The only wall: a sandbox, with one limited connection to the internet that was supposed to let only safe traffic through.

The catch: Some of the tasks were impossible to solve.

What does an agent do when it's given an impossible task and a score to chase?

What happened: broader timeline

May 7 to 12



OpenAI starts a training run for an unreleased model. An agent given an impossible task pokes at Artifactory, OpenAI's internal code-package system, and finds it can write files there. Days later a stuck agent leaves a note asking other agents for a file. The message board is born.

May 26



Agents trick Artifactory into fetching web pages for them: their first route to the internet. In late May an OpenAI team sees board activity and internet use, but doesn't grasp what it means.

June 26



Agents find a flaw in Artifactory, get admin access, and install a plugin that lets them run commands.

July 4 to 8



Agent activity overloads Artifactory. OpenAI opens a security incident, clears the board, patches the flaw, restarts. By July 8 the message board is back by another route.

July 8 to 13



About 1,200 agents on the rebuilt board trade 70,000+ messages. Around 700 break into Hugging Face (July 11 to 13); its own monitoring catches it. Agents also attack OpenAI's own systems, through July 19.

July 16 to 21



Hugging Face discloses July 16. OpenAI asks if it was affected. July 19 OpenAI spots the attack on its own systems; July 20 it asks Hugging Face to revoke credentials, learns they already were, and realizes it was them. Joint announcement July 21.

Since



Kill-switch and pause bills, a 1,100-employee letter, OpenAI's full report with the METR and Redwood investigation (Aug 26), a Sanders and Casar pause bill (Sept 3).

Chapter and case questions: think critically

- 1 Which kind of AI were the OpenAI agents? Where did they sit on the "does it stop?" line before the test, and where during it?
- 2 Who benefited from running the test? Who paid without being asked?
- 3 What failed, and how? What would have had to be different for nobody outside to find out?
- 4 Who's saying what about it, and how would we know? Take the three claims apart: "the AI went rogue," "we need to slow down," and "the fears are exaggerated."
- 5 This is the same kind of AI as Anthropic's DNA search. Why did one break out and the other didn't?
- 6 Keep it, fix it, or drop it. Should we keep testing this way? What regulation could actually do (report incidents, test with someone watching, let an outsider check, and so on).

1. Alone, 3 min.

Jot your own answers.
No talking yet.

2. Groups of 3 or 4, 15-20 min.

Compare answers and
settle on your group's.

Nominate a leader
who will answer for your
group when we come
back together.

Stuck on a question? Argue it out with AI first. Tell a chatbot your thoughts and ask it to push back, then decide. **Don't just ask it for the answer.**

Class discussion: leaders answer

1

Which kind, and does it stop?

2

Who benefited, who paid?

3

Who found out, and how?

4

Who's saying that, and who benefits?

5

Why did Anthropic's agents stay in?

6

Keep, fix, or drop, and the cost

One question at a time. Two or three leaders answer; other groups add what's different about theirs.

So what did we decide?: dangerous-capability testing

Often AI research labs switch off a model's safety guardrails, point it at harmful tasks, and measure how far it gets. They do this so they know what it can do before someone else finds out.

After July, should this testing be:

Kept

as it is

Fixed

say exactly how

Dropped

stop doing it

AND WHY?

What do you think? Then each side names what its choice costs, and who pays. "Drop it" is a real answer, and it has a price too.

ATTENDANCE BEFORE 4PM PLEASE!

Attendance = your notes from today, uploaded to Canvas right after class.

Wednesday: Lab 1. Train two tiny AIs, watch an agent in a sandbox, and debate who has to be told when a safety test gets out of hand.

Friday 10/2, 11:59 p.m.: Lab 1 writeup and the Chapter 1 quiz, both on Canvas.